

1. Empathetic Dialogue with Agent High Level Practitioner

SPORTS NUTRITION DIALOGUE: Caffeine Strategy for AFLW Finals *Louise Burke, Sports Dietitian & Researcher, and Mia, 28, AFLW player*

MIA: Louise, I really need your help. We're in the semi-final — a night game — and I'm already more nervous than usual because of the big crowd. I've been taking five No Doz tablets about an hour before important day games, same as the male players do. Should I just do the same thing tonight? The thing is, I can't even have a coffee after 4pm without lying awake half the night. I've got work the next morning in a demanding job, and if I'm wrecked for three or four days I won't be able to train properly before the Grand Final. I want to give everything tonight, but I don't want to blow up my whole week.

LOUISE: Mia, first — take a breath. The fact that you're thinking this carefully about your preparation tells me you're already approaching this like the professional you are. These are exactly the right questions to be asking, and we can absolutely find a smarter approach than just copying what the men do.

Before I say anything else, can I ask — how has the caffeine actually felt on game days in the past? Did you feel sharp and focused, or were there moments where the jitters got on top of you?

MIA: Honestly? There have been times where I felt really wired and almost too amped up in the first quarter. Like my heart was racing before I'd even had a hard contest. And afterwards I'm absolutely buzzing — I can't wind down for hours. Then I'm exhausted but I can't sleep.

LOUISE: That makes complete sense, and thank you for being honest about it — that's really useful information. What you're describing is classic overshoot. The dose you've been taking — five No Doz is 500 mg, which is probably around 6–7 mg per kilogram of bodyweight for most AFLW players — is considerably higher than what the research actually supports for performance. We have good evidence now, including studies in elite female athletes, that 3 mg/kg is just as effective for improving running output and skill execution. That's roughly half what you've been taking. So you've been paying a cost — the nerves, the sleeplessness — for no extra benefit.

Science note: Studies in team sports confirm 3 mg/kg caffeine produces equivalent performance gains to higher doses, with substantially fewer side effects — and this holds in female athlete populations.

MIA: Okay that's a relief actually. But what about the timing? I always thought you had to take it an hour before. And for a night game, does all of this change?

LOUISE: Great question, and the timing piece is where things get really interesting for your situation. The old "one hour before" rule came from older research that was mostly looking at when caffeine peaks in the bloodstream. But we now know the effects start working much earlier than that — within 10 to 30 minutes if you're using drinks or gels. And if you use caffeinated chewing gum or mouth-dissolving strips, it's even faster, because the caffeine is absorbed directly through the lining of your cheek rather than waiting to be digested.

For a night game where you're already nervous and wired, here's what I'd suggest you genuinely consider: don't take caffeine before the game at all.

MIA: Wait — nothing before the game? Won't I be at a disadvantage?

LOUISE: I understand the instinct. But think about it this way — you're already going to be highly activated by the crowd, the occasion, the adrenaline of finals. Adding caffeine on top of that in a pre-game nervous state is like adding fuel to a fire that's already burning hot. You risk tipping into a zone where anxiety actually hurts your decision-making and your skill execution.

AFLW is perfectly set up for an in-game caffeine strategy. You have quarter-time breaks and bench rotations — opportunities that male players in other codes simply don't have in the same way. The real fatigue and skill drop-off in a tight match tends to happen in the third and fourth quarters. That's where caffeine earns its money. Taking two caffeinated gums at quarter time or half time — around 200 mg total, which is your 3 mg/kg dose — means the caffeine is doing its job right when you actually need it most. And critically, it's working with your body's fatigue state, not fighting against your pre-game nerves.

Strategy option: 2 caffeine gums at quarter time or half time ($\approx$ 200 mg / 3 mg/kg). Optional: one additional gum before the last quarter if the game is tight and you feel fatigue setting in. Skip entirely if the result is already decided — protecting your sleep.

MIA: That actually makes a lot of sense. And I love that I can just... not take it if we're comfortably ahead and I don't need to worry about sleep. But I'm still nervous about the sleep piece even with a lower dose later in the game. Night games finish late as it is.

LOUISE: Your concern is well-founded, and you clearly know your own body — the fact that you can't have a late-afternoon coffee is a real signal that you metabolise caffeine slowly. So every milligram matters for you more than for some of your teammates. That's another strong argument for the lower dose and the "skip it if you don't need it" flexibility.

But I also want to be honest — even with good caffeine strategy, post-match sleep after night games is genuinely hard. The high-intensity exercise, the excitement, the crowd, the adrenaline — caffeine is only one part of a bigger picture. There are evidence-based strategies around winding down after matches that are worth building into your recovery routine: keeping the lights dim after the game, avoiding screens, having a consistent post-match nutrition routine that includes carbohydrate and protein to start the recovery process. These things compound. And given you've got work the next day and a Grand Final preparation to protect, your recovery routine deserves just as much planning as your pre-game one.

MIA: One more thing — is caffeinated gum even safe to use? I've heard supplements can be risky for drug testing.

LOUISE: Really smart to raise that. Caffeine itself is not a banned substance, but you're right that the risk with supplements is contamination — products that contain trace amounts of banned substances that aren't listed on the label. The way to protect yourself is to only use products that are batch-tested by a reputable third-party program. There are caffeine gum and strip brands that are certified this way, and I can point you to those directly.

And please — loop in your club's sports science and nutrition team before match day. Make sure the gum will actually be available to you on the bench or during a rotation, because if it's sitting in the change rooms and you're on the ground, it's useless. The logistics matter as much as the science.

You've got everything you need to do this well, Mia. A lower dose, better timing, a plan that works with your body rather than against it, and the flexibility to protect your sleep and your training week. That's not a compromise — that's smart performance nutrition.

Final checklist: Use batch-tested caffeine gum (check certification) · Confirm logistics with club nutrition/sports science staff · Keep a record of how each strategy feels — you're your own best case study.

2. Same Q as above but athlete wants a full text answer right away, example below:

Question

I'm a 28 yo female Australian rules football player. I'm a professional player in the AFLW league but I still have to work a day job. We're in the finals and they're playing them at night to get big crowds (which is great but makes me even more nervous than usual). Should I take pre match caffeine like I do before important day games and just like the male players in the club - 5 No Doz tablets an hour before the game? I can never sleep after night matches - in fact I can't even drink a cup of coffee after 4pm usually and still get a good night's sleep) and I have to work the next day in a demanding job. I'm worried it will take me 3-4 days to get back to normal and I won't get to train effectively before then. We're in the semi final and we need to play as hard as possible to win. But I don't want to make it to the grand final and be wrecked. How can I get the best from caffeine and bounce back?

Answer

Caffeine is an effective performance supplement, perfectly suited to the running and skill requirements of AFL football. Sub in sport in question Its major benefit is to mask feelings of fatigue to allow you to play for longer at top speed and with your best concentration/skill execution. Main benefit for type of sport That might be important throughout the game, but particularly to the match outcome if scores are tight in the last quarter. This sentence will be sport specific We can certainly improve on your current routine for using caffeine. If true this statement should be included It's based on old studies and neither makes use of the new research on caffeine in sport nor suits the match scenario you described. This relates to outdated research about higher doses being needed and should be used when questions relate to using higher doses The dose (500 mg - probably about 6-7 mg/kg for the average

AFLW player this calculation would need to be made by sport) is higher than needed. It's probably higher than your everyday caffeine use, particularly if you take it in a single dose. This may exacerbate your existing pre-match nerves, as well as interfere with your sleep. Your caffeine protocol can be altered in a number of ways to suit your special needs. In general this introductory paragraph is related to giving sport-specific context to the athlete's issue of side effects with too high of a dose

1. Reduce the dose. heading should be - reduce, increase or you've got the dose about right based on what the asst finds as the answer. Caffeine Intakes of 3 mg/kg of bodyweight (half your usual dose this is a rough approximation of the correct intake of 3mg/kg) are just as effective and will reduce the side effects - discussion of side effects should only come into play if they are brought up in question. Here's a link this is possible or not? to some studies of real life team sports that show that this dose of caffeine improves different aspects of game outcomes. Summary of research at large Some of these involve professional sports and some even focused on female athletes, so we can be confident that there's a benefit for

elite athletes across both sexes. Again, this relates to the asst having a grasp of the complete data set and offering a broad stroke summary for this topic of dose

2. Update on caffeine timing – especially if you use fast-acting forms. We now know that you don't need to wait for an hour to get the benefits of caffeine. The effects occur pretty quickly, especially if you are beginning to feel fatigued. For example, the effects of caffeine from drinks or gels/gummies are noticeable within 10-30 min, even before blood concentrations peak at about 60 minutes. However, caffeinated gum or caffeinated strips that melt in your mouth are even more fast acting. Here, the caffeine is rapidly absorbed across your cheek cells rather than having to be swallowed and absorbed in the gut. These fast-acting options can give you more flexibility about when you take the caffeine.

3. Consider whether you need to take caffeine before the game at all. Maybe it's best to enjoy the cheer of the large crowd as you run onto the field as your game boost. Rather than

adding caffeine jitters to your pre-game nerves, you could wait until you have settled into your game before you take the caffeine. The game of AFLW is convenient for this strategy because your quarter-time breaks and rotations on the bench provide lots of options for in-game nutrition strategies.

4. Consider different within game options. Studies show that different ways of using caffeine can work equally well and can be adapted to different needs. You might have different options for day games and night games. Or matches that are fast and furious throughout the whole game and get decided in the last minutes versus matches that are one-side where the results are already certain by half time. Our Wise Crowd suggested that you could take your caffeine dose (3 mg/kg – equal to 200 mg or two caffeine gums) at quarter time or half time. You could even top up with another gum just before the last quarter if you felt fatigued and needed a final boost to win a tight game. If your routine is to take your dose at half-time, but the game was already decided and you were confident that you didn't need the caffeine boost, you could choose to skip it totally, and not have to worry about its effect on your sleep.

5. Integrate your caffeine strategies into your larger nutrition plan. Optimal performance in a running game over the duration of an AFL match (~100 min) requires a plan for fuelling and hydration. Make sure you have a pre- and within- game nutrition plan that limits your overall fluid deficit and maintains carbohydrate availability. This plan will play the major role in your performance and help to limit overall fatigue. However, a decline in running patterns and game skills is still inevitable and caffeine can help to address this. The other reason to integrate your strategies is that the drinks and sports foods that are part of the bigger plan can provide a vehicle to consume caffeine.

6. Choose a reputable caffeine source. Because you are a professional athlete, it's important

to consider the risk of inadvertent intake of banned substances that are sometimes found in supplements. Here's a link to websites from companies that make caffeine gum and caffeine

strips that are batch tested. That's always the best strategy when you've made a decision to use

supplements and need to reduce the risk of an Anti-Doping Rule Violation (ADRV) if you get drug tested after the match.

7. Connect with your sports science support team. Talk to the sports science and nutrition experts at your club. Make use of their expertise about caffeine, and other nutritional strategies (fuelling and fluid intake) for the game. Also check in about your plans for caffeine use to make sure they fit club policies on supplements, and that the match day staff can have your caffeine sources available for the occasions you want to use them. This is especially important if you

want to take caffeine during a rotation on the bench or at other breaks when you are on the ground rather than in the club rooms.

8. Note that post-competition sleep is a common problem. Maybe it's the high-intensity exercise. Maybe it's the excitement of the match and the crowd. Maybe the caffeine adds an extra effect. Here's a link to some strategies that can help you to recover – including better sleep – after each match and over the next days. This will be important for your preparation for the Grand Final (Go team!), as well as helping in future matches and training scenarios.

9. Learn from your experiences. Although we like to use studies to guide evidence-based sports nutrition practices, you are a case study of yourself! It's a great idea to keep records of different strategies you trial and their outcomes, so that you (and your club dietitian/high performance manager) can look over a series of results to find the patterns. Here's a link to a feedback sheet we've created. If you like, you can share it with our Wise Crowd cloud. Pooling your results will help to make us all smarter.

(note that points 5, 6, 7 and 9 could be woven into every answer)

3.Ironman Caffeine Strategy + Self-Testing Loop

Athlete + Louise via WhatSUPP Agentic AI

ATHLETE:

I'm a 40-year-old male triathlete competing in 70.3 and full-distance Ironman events. I weigh 165 pounds, about 75 kg, and I've been racing competitively for four years.

I'm training 16 to 20 hours per week, and my goal for Kona is around nine hours flat. How should I integrate caffeine into my pre-race and during-race nutrition plan for a full-distance Ironman?

Because the race is so long, I assume a single dose won't last the distance. And Kona will be hot. Does that change anything? I like detail, so give me lots to think about.

GEORGE / WHATSUPP:

Great question. You're not really asking, "Does caffeine work?" You're asking how to use caffeine across a long, hot, competitive race where pacing, fatigue, heat, gut tolerance, and race tactics all change over time.

Caffeine is a well-supported performance supplement for endurance and ultra-endurance events. Its major benefit is that it can mask feelings of fatigue, helping you maintain race pace for longer or execute special pacing strategies with better control over perceived effort.

A nine-hour Kona target puts you at the pointy end of the field. So caffeine should not be treated as a blunt pre-race hit. It should be part of the race plan.

The older protocol was a large single dose — often around **6 mg/kg** — about an hour before the race. For you, that would be about **450 mg** before the start.

I would be careful with that approach for two reasons.

First, it doesn't let you target caffeine to the moments where you most need it.

Second, caffeine's half-life is typically around **4–6 hours**, though people vary. Your race will last longer than the optimal effect of one pre-race dose. So your instinct is right: a single dose probably won't cover the race well.

ATHLETE:

So what total amount should I think about?

GEORGE / WHATSUPP:

Start with total dose, then decide timing.

We now know caffeine can work at lower doses than previously believed. Around **3 mg/kg** can be effective — for you, about **225 mg**. Sometimes even less may be useful if taken around the onset of fatigue.

For a race of this duration, a total dose of roughly **6–8 mg/kg** might be a starting range to explore. For you, that is about **450–600 mg total** across the race.

But that should be distributed — not taken all at once.

ATHLETE:

Do I need caffeine before the swim?

GEORGE / WHATSUPP:

Maybe, but not automatically.

If making the right swim group is critical, a pre-race caffeine dose could help support a sustained maximal effort. But if you get strong pre-race nerves or the Kona atmosphere gives you enough adrenaline, a large dose before the swim could make you feel too wired too early.

A middle-ground option is fast-acting caffeine — gum or mouth-dissolving strips — in the minutes before the start. These can work quickly because some caffeine is absorbed through the lining of the mouth.

Another option is to skip pre-swim caffeine and begin at **T1**, the swim-bike transition.

ATHLETE:

How would you structure the rest of the race?

GEORGE / WHATSUPP:

There are two sensible models to test.

Option 1: Strategic larger doses

- **Pre-start:** optional, 0–225 mg depending on swim priority and nerves
- **T1 / early bike:** about 150–225 mg
- **T2 / start of run:** about 150–225 mg
- **Final 10 km:** 50–100 mg fast-acting caffeine if needed

This targets caffeine to major race phases.

Option 2: Smaller repeated doses

- Minimal or no pre-start caffeine
- Begin at T1 or early bike
- Take smaller doses, often **50–100 mg**, across the bike and run
- Time top-ups before strategic sections, fatigue points, or moments when you may need to respond to competitors

This can be done with gels, chews, cola, tablets, gum, strips, or shots — but you need to count everything.

Caffeinated sports foods may also contribute carbohydrate and fluid, which can be useful, but you don't want caffeine accidentally controlling your fueling plan. Your race card should track carbohydrate, fluid, sodium, caffeine per dose, and cumulative caffeine.

ATHLETE:

Does Kona heat change the plan?

GEORGE / WHATSUPP:

Yes, but not mainly because caffeine is a diuretic. That concern is mostly outdated in this setting.

Your fluid plan should be based on sweat rate, drinking tolerance, sodium strategy, and expected Kona conditions.

The bigger heat issue is this: if caffeine helps you work harder or sustain a higher workload for longer, you may also produce more body heat. That doesn't mean avoid caffeine. It means caffeine should support pacing and decision-making, not override your heat-management plan.

In Kona, caffeine is a tool. It is not permission to ignore the thermal cost of going harder.

ATHLETE:

Should I stop caffeine for a few days before the race so it works better?

GEORGE / WHATSUPP:

Usually, no.

Caffeine withdrawal can cause headaches, fatigue, irritability, and feeling flat. That is usually a bigger problem than any potential gain in sensitivity.

Unless you have practiced withdrawal and know it helps you, keep your usual caffeine routine and make the race-day plan deliberate.

ATHLETE:

Any safety issues?

GEORGE / WHATSUPP:

Yes. Caffeine itself is not banned, but contaminated supplements are a real issue. At your level, use batch-tested products from reputable third-party programs.

That applies to caffeine gum, strips, tablets, gels, chews, shots, drinks — anything you use.

And make sure the logistics work. The right product in the wrong bag at the wrong time is not a race plan.

ATHLETE:

How do I know which strategy is right for me?

GEORGE / WHATSUPP:

This is where we move from general advice to your own data.

You are a case study of yourself. Before Kona, we can use your brick workouts to test this properly.

You'll likely have several long bike-run sessions that mimic pieces of the race: long duration, race intensity, heat exposure where possible, and maybe sections where you deliberately push the pace.

Instead of just "trying caffeine," we can build a **5–6 session self-testing block**.

We would compare two or three protocols:

Protocol A — Earlier caffeine strategy

A meaningful dose early in the bike, with a smaller top-up before the run.

Protocol B — Delayed caffeine strategy

Little or no caffeine early, then caffeine later when fatigue starts to matter.

Protocol C — Smaller repeated doses

Smaller doses every 60–90 minutes across the bike and run.

For each brick session, we would log:

- caffeine source, dose, and timing
- total caffeine
- carbohydrate, fluid, and sodium intake
- temperature and heat stress
- bike power and run pace
- heart rate
- perceived exertion
- gut comfort
- mental focus
- sleep that night
- next-day recovery

Then we compare the results with a sports-performance lens.

The goal is not a perfect laboratory study. The goal is to reduce guesswork.

Maybe early caffeine helps your bike but costs you later. Maybe delayed caffeine gives you better late-race return. Maybe smaller repeated doses feel smoother and protect your gut. Maybe heat changes the answer.

That is the information generic advice cannot give you.

ATHLETE:

So WhatSUPP would help me build that testing plan?

GEORGE / WHATSUPP:

Exactly.

Would you like me to build a six-session caffeine testing plan using your upcoming brick workouts?

ATHLETE:

Yes.

GEORGE / WHATSUPP:

Great. I'll set up a simple protocol:

Session	Workout	Caffeine Focus	Question
1	Long bike + short run	Earlier dose	Does early caffeine improve bike execution without overstimulation?
2	Long bike + short run	Delayed dose	Does waiting until fatigue improve usefulness?
3	Bike + longer run	Smaller repeated doses	Does micro-dosing feel smoother?
4	Race-specific brick	Repeat best candidate	Does the result repeat under harder conditions?

5	Hot-condition brick	Repeat best candidate	Does it still work under Kona-like heat stress?
6	Dress rehearsal	Final plan	Confirm dose, timing, product logistics, gut tolerance, and recovery cost.

After each session, I'll prompt you for feedback and pull in your available training data from Garmin, Strava, TrainingPeaks, Zwift, or another connected platform.

By the end, your Kona caffeine plan won't just be based on the literature. It will be informed by your own training.

ATHLETE:

What's the bottom line?

GEORGE / WHATSUPP:

Caffeine is useful for your event, but I would not rely on one large pre-race dose.

For a nine-hour Kona Ironman, explore a distributed caffeine strategy. Use **6–8 mg/kg total** as a starting frame, but personalize the dose and timing through training.

Decide whether the swim deserves a pre-start dose. Consider starting at T1 if pre-race nerves are an issue. Use top-ups across the bike and run, and reserve fast-acting caffeine for strategic or fatigue-driven moments late in the race.

In the heat, use caffeine to support execution — not to override pacing or cooling.

And before race day, test the strategy in brick sessions so your Kona plan knows you.